20 GENES/PROTEINS ASSOCIATED WITH PROSTATE CANCER

S/N	Gene Symbol	Protein Name	Cancer Type	PDB Structure Available	PDB ID	References
1	KLK3	Prostate-Specific Antigen (PSA)	Prostate Cancer	Yes	9ZJQ | pdb_00009zjq	To be published.
2	AR	Androgen Receptor		Yes	8FGY | pdb_00008fgy	Doamekpor SK, Peng P, Xu R, Ma L, Tong Y, Tong L. (2023). A partially open conformation of an androgen receptor ligand-binding domain with drug-resistance mutations. *Acta Crystallogr F Struct Biol Commun.*
3	PTEN	Phosphatase and Tensin Homolog		Yes	7JTX | pdb_00007jtx	Dempsey DR, Viennet T, Iwase R, Park E, Henriquez S, Chen Z, Jeliazkov JR, Palanski BA, et. al., (2021). The structural basis of PTEN regulation by multi-site phosphorylation. *Nat*

						Struct Mol Biol.
4	TP53	Cellular Tumor Antigen		Yes	[8DC4](#) | [pdb_00008dc4](#)	Guiley KZ, Shokat KM. A (2023). Small Molecule Reacts with the p53 Somatic Mutant Y220C to Rescue Wild-type Thermal Stability. *Cancer Discov.*
5	AMACR	Alpha-methylacyl-CoA racemase		Yes	[8RP3](#) | [pdb_00008rp3](#)	Mojanaga OO, Woodman TJ, Lloyd MD, Acharya KR. (2024). α-Methylacyl-CoA Racemase from *Mycobacterium tuberculosis*-Detailed Kinetic and Structural Characterization of the Active Site. *Biomolecules.*
6	EZH2	Enhancer of Zeste Homolog 2		Yes	[4MI0](#) | [pdb_00004mi0](#)	Wu H, Zeng H, Dong A, Li F, He H, Senisterra G, Seitova A, Duan S, Brown PJ, Vedadi M, Arrowsmith CH, Schapira M. (2013). Structure of the catalytic domain of

						EZH2 reveals conformational plasticity in cofactor and substrate binding sites and explains oncogenic mutations. *PLoS One.*
7	TOP2A	Topoisomerase IIα		Yes	**1ZXM** | **pdb_00001zxm**	Wei H, Ruthenburg AJ, Bechis SK, Verdine GL. (2005). Nucleotide-dependent domain movement in the ATPase domain of a human type IIA DNA topoisomerase. J *Biol Chem.*
8	MKI67	Ki-67 Antigen		Yes	**5J28** | **pdb_00005j28**	Kumar GS, Gokhan E, De Munter S, Bollen M, Vagnarelli P, Peti W, Page R. (2016). The Ki-67 and RepoMan mitotic phosphatases assemble via an identical, yet novel mechanism. *Elife.*
9	CCNB1	Cyclin B1		Yes	**4Y72** | **pdb_00004y72**	Brown NR, Korolchuk S, Martin MP, Stanley WA,

						Moukhametzianov R, Noble MEM, Endicott JA. (2015). CDK1 structures reveal conserved and unique features of the essential cell cycle CDK. *Nat Commun.*
10	RRM2	Ribonucleotide reductase regulatory subunit M2		Yes	2JRS | pdb_00002jrs	To be published.
11	CKS2	Cyclin kinase subunit 2		Yes	4YC3 | pdb_00004yc3	Brown NR, Korolchuk S, Martin MP, Stanley WA, Moukhametzianov R, Noble MEM, Endicott JA. (2015). CDK1 structures reveal conserved and unique features of the essential cell cycle CDK. *Nat Commun.*
12	TK1	Thymidine kinase 1		Yes	2JA1 | pdb_00002ja1	Kosinska U, Carnrot C, Sandrini MP, Clausen AR, Wang L, Piskur J, Eriksson S, Eklund H. (2007). Structural studies of thymidine kinases from Bacillus

						anthracis and Bacillus cereus provide insights into quaternary structure and conformational changes upon substrate binding. *FEBS J.*
13	BIRC5	Survivin		Yes	**1XOX** | **pdb_00001xox**	Sun C, Nettesheim D, Liu Z, Olejniczak ET. (2025). Solution structure of human survivin and its binding interface with Smac/Diablo. *Biochemistry.*
14	PSCA	Prostate stem cell antigen		Yes	**9U9N** | **pdb_00009u9n**	Shulepko MA, Che Y, Paramonov AS, Kocharovskaya MV, Kulbatskii DS, et.al., (2025). Pro-Inflammatory Protein PSCA Is Upregulated in Neurological Diseases and Targets β2-Subunit-Containing nAChRs. *Biomolecules.*
15	KLK2	Kallikrein-Related Peptidase 2		Yes	**4NFE** |	Skala W,

					pdb_00004nfe	Utzschneider DT, Magdolen V, Debela M, Guo S, Craik CS, Brandstetter H, Goettig P, et. al., (2014). Structure-function analyses of human kallikrein-related peptidase 2 establish the 99-loop as master regulator of activity. J *Biol Chem.*
17	FOXA1	Forkhead Box Protein A1		Yes	8VFZ | pdb_00008vfz	Zhou BR, Feng H, Huang F, Zhu I, Portillo-Ledesma S, Shi D, et. al., (2024). Structural insights into the cooperative nucleosome recognition and chromatin opening by FOXA1 and GATA4. *Mol Cell.*
18	STEAP2	Six Transmembrane Epithelial Antigen of the Prostate 2		Yes	7TAI | pdb_00007tai	Chen K, Wang L, Shen J, Tsai AL, Zhou M, Wu G. (2023). Mechanism of stepwise electron transfer in six-transmembrane

						epithelial antigen of the prostate (STEAP) 1 and 2. *Elife*.
19	SLC45A3	Prostein		Yes	4CQE | pdb_00004cqe	Pulici M, Traquandi G, Marchionni C, Modugno M, et. al., (2015). Optimization of diarylthiazole B-raf inhibitors: identification of a compound endowed with high oral antitumor activity, mitigated hERG inhibition, and low paradoxical effect. *ChemMedChem*.
20	WNT5A	Wnt Family Member 5A		Yes	9FSE | pdb_00009fse	Griffiths SC, Tan J, Wagner A, et.al. (2024). Structure and function of the ROR2 cysteine-rich domain in vertebrate noncanonical WNT5A signaling. *Elife*.

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